

CAR RENTAL REQUEST AUTOMATION IN SERVICENOW

Phase 3: Project Design Phase

Describes how the solution is structured: the process, the data, the screens, the automation, and the decisions behind them.

Item	Details
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Platform	ServiceNow (Service Catalog, Flow Designer, Business Rules)
Document	Phase 3 of 5

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3.1 Design Goals and Principles

- **Keep intake structured.** Every value the Transport team needs is a separate field, and the only list (car type) is a controlled choice.
- **Keep decisions on the record.** Approval is written to the request itself, not to an email thread.
- **Automate hand-offs.** No step depends on a person remembering to forward or notify.
- **Separate the normal path from exceptions.** Approval and fulfilment live in the flow; unavailable cars are handled by a business rule.
- **Stay on platform.** Use standard ServiceNow objects (REQ, RITM, sc_task) so the solution is familiar to any ServiceNow admin.

3.2 Solution Architecture

The solution has four layers that hand work to one another:

Layer	Component	Responsibility
Presentation	Service Catalog (Transportation Services category)	Shows the Car Rental Request form and the Order Now button
Data	REQ (sc_request), RITM (sc_req_item), catalog variables	Stores the order and the values the user entered
Automation	Flow Designer flow "Car Rental Req"	Approval, condition, task creation, notification
Fulfilment / Exceptions	Catalog Task (sc_task), Business Rule, Incident	Transport work and unavailable-car handling

3.3 Process Flow

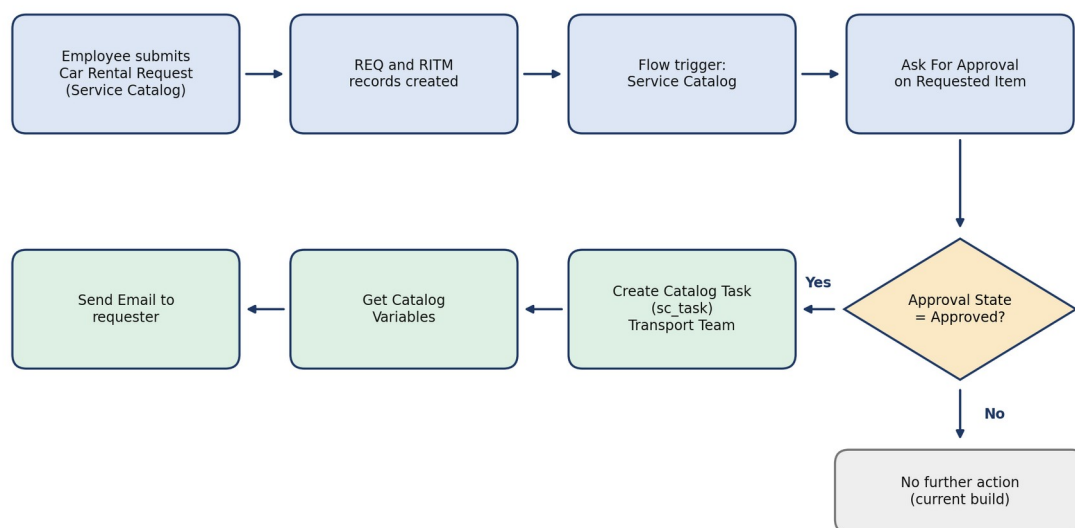


Figure 3.1: End-to-end process flow of the car rental automation

The top row is the path every request takes up to the approval. The bottom row runs only when the approval outcome is Approved. A rejected request ends at the decision in the current build.

3.4 Data Architecture

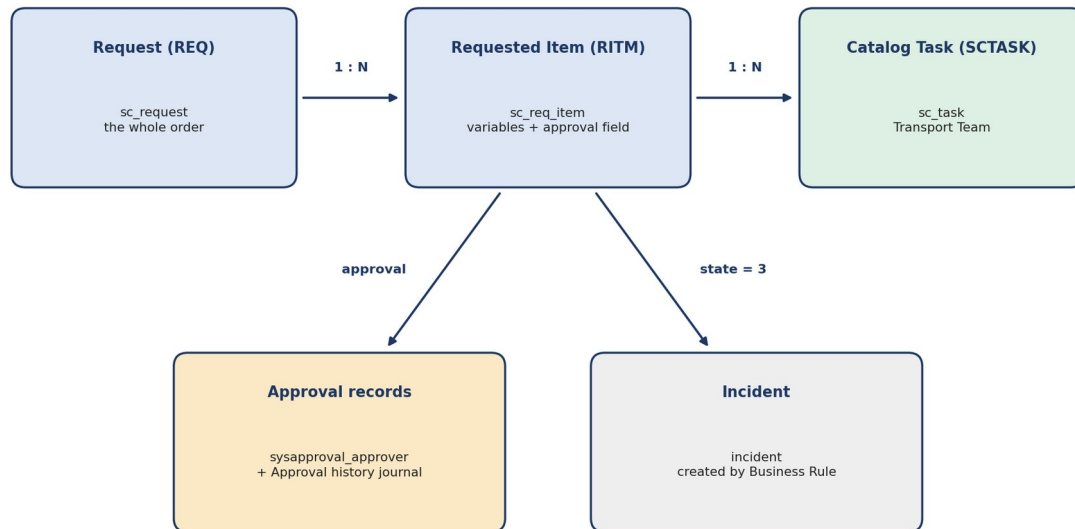


Figure 3.2: How the request records relate to one another

Table / Object	Purpose
sc_request (REQ)	Parent record for the whole order
sc_req_item (RITM)	The car rental item. Holds the catalog variables and the approval field
sc_task (SCTASK)	Fulfilment task assigned to the Transport Team
Approval records and Approval history journal	Track request progress and the decision history
incident	Receives an incident when a car is reported unavailable (Business Rule)
Car Inventory table (design scope)	Would hold car availability and assignment details, a planned extension

Concept note: REQ, RITM, SCTASK. Every catalog order creates one **REQ**, which is the order or cart. Each item in it becomes a **RITM**, and the variables the user filled in live on the RITM. Fulfilment work is broken into one or more **SCTASKs** under that RITM. This is why the flow's trigger hands over the *Requested Item Record*: it carries both the form data and the approval state that the rest of the flow depends on.

Where each variable is stored and used

Variable	Stored on	Used by
requestor	RITM	Send Email: the To address comes from the requestor's email
requested_date	RITM	Business Rule: quoted in the incident description

Variable	Stored on	Used by
pickup_location, drop_location, duration_hrs, car_type, reason	RITM	Available to the flow through Get Catalog Variables; not yet referenced in the email body or task text

3.5 Catalog Form Design

The form is short and unambiguous. Each field maps to one variable, and Car Type is a controlled list so the Transport team never has to interpret free text. The catalog item sits in the **Service Catalog**, category **Transportation Services**, with the short description "Request a car for rental".

Order	Label	Control	Values / format	Purpose
1	Car Type	Dropdown	Mini, SUV, Sedan	Tells Transport which class of car to arrange
2	Requested Date	Date picker	Calendar date	When the car is needed
3	Pickup Location	Text box	Free text	Where the trip starts
4	Duration (Hours)	Radio buttons	0, 1, 2, 3, 4, 5	How long the car is needed
5	Requestor	User lookup	A user record	Who the request is for, and who receives the email
6	Drop Location	Text box	Free text	Where the trip ends
7	Reason	Multi-line text	Free text	Business justification for the approver

On the right of the form, the **Order this Item** panel shows Quantity (1), Delivery time (0 Days), and the **Order Now** and **Add to Cart** buttons.

The screenshot shows the ServiceNow interface for a 'Car Rental Request' catalog item. The form is titled 'Request a car for rental' and contains the following fields:

- Car Type:** A dropdown menu with 'SUV' selected.
- Requested Date:** A date picker field.
- Pickup Location:** A text input field.
- Duration (Hours):** Radio buttons for 0, 1, 2, 3, 4, and 5. The '0' option is selected.
- Requestor:** A user lookup field with a magnifying glass icon.
- Drop Location:** A text input field.
- Reason:** A multi-line text area.

On the right side, the 'Order this Item' panel displays:

- Quantity:** 1 (with a dropdown arrow).
- Delivery time:** 0 Days.
- Buttons:** 'Order Now' (blue) and 'Add to Cart' (light blue).
- Shopping Cart:** A section showing 'Empty'.

Figure 3.3: Car Rental Request form as seen by the requester

3.6 Flow Design

The flow is triggered by the Service Catalog and has five steps. The structure below is the flow as it appears in Flow Designer.

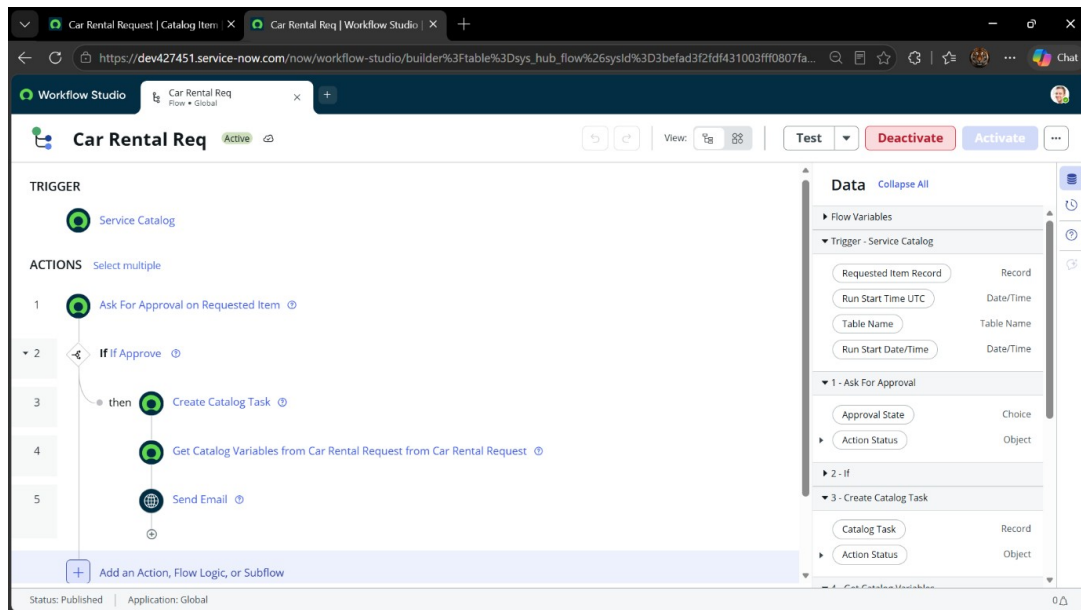


Figure 3.4: Structure of the Car Rental Req flow (trigger and five actions)

Step	Action	Key inputs	Output used later
Trigger	Service Catalog	None	Requested Item Record
1	Ask For Approval	Record = Requested Item (sc_req_item); Approval Field = Approval; Journal Field = Approval history; Rule = Approve when Anyone approves	Approval State
2	If ("If Approve")	Approval State is Approved	Decides whether steps 3 to 5 run
3	Create Catalog Task	Table = sc_task; Requested Item; Assignment group = Transport Team; Description; Wait enabled	Catalog Task record
4	Get Catalog Variables	Submitted Request = Requested Item; template = Car Rental Request	requestor, requested_date, and the other variables
5	Send Email	Target Record = Requested Item; To = requestor's email; Subject; Body; Include Watermark	None

3.7 Business Rule Design

Property	Value
Name	Create Incident If Car Unavailable

Property	Value
Table	Requested Item (sc_req_item)
When to run	After; on Insert and Update (not on Delete or Query)
Order	100
Type	Advanced (uses a script); application scope Global
Logic	If the record's state equals 3, insert an incident
Incident content	Short description "Car unavailable for request: " plus the request number; description "Car not available for trip on " plus the requested date

The rule runs **after** the database operation, so the RITM is already saved when the incident is created. It reads the requested date through `current.variables`, which is how a business rule accesses catalog variables.

3.8 Notification Design

Property	Value
Trigger point	After the catalog task is completed (the Create Catalog Task action waits)
Recipient	The email address of the Requestor variable
Subject	Your Car Request is approved.
Body	"Dear, Your car Rental Request is approved! Regards."
Watermark	Included, so replies can be linked back to the record

3.9 User Journeys by Role

Role	Journey
Requester	Open the item, fill in the form, click Order Now, note the REQ number, and follow the request record
Approver	Open the RITM, find the approval, and approve or reject
Transport Team	Receive the catalog task for the group, arrange the car, and close the task
Admin	Maintain the catalog item and variables, the flow, and the business rule; move the Update Set

3.10 Usability, Help Text and Tooltips

- Every variable has a clear question label, and related fields (pickup and drop) use parallel wording.
- Car Type and Duration use fixed options to prevent typing errors.

- Help text and tooltips should say what value is expected, for example "Duration in hours" and "Enter the city or address".
- The form fits one screen with the Order Now button always visible on the right.

3.11 Security and Access Design

- Approval is tied to a named approver in the rule, and the decision is stored on the RITM.
- Tasks are assigned to a group, so access follows group membership rather than a named person.
- The flow runs as the **System User**, so it can create the task and send the email regardless of the requester's own roles.

3.12 Key Design Decisions

Decision	Reason
Trigger = Service Catalog	Fires when the catalog item is ordered and gives the flow the RITM and its variables
Approval written to the Approval field and Approval history journal	Preserves the decision trail on the record itself
Task table = Catalog Task (`sc_task`)	The standard fulfilment object in the catalog lifecycle
Assign to a group , not a person	The Transport Team can change members without editing the flow
Unavailable cars handled by a Business Rule	Keeps exception handling separate from the main approval path
Flow runs as System User	Avoids permission failures for ordinary requesters